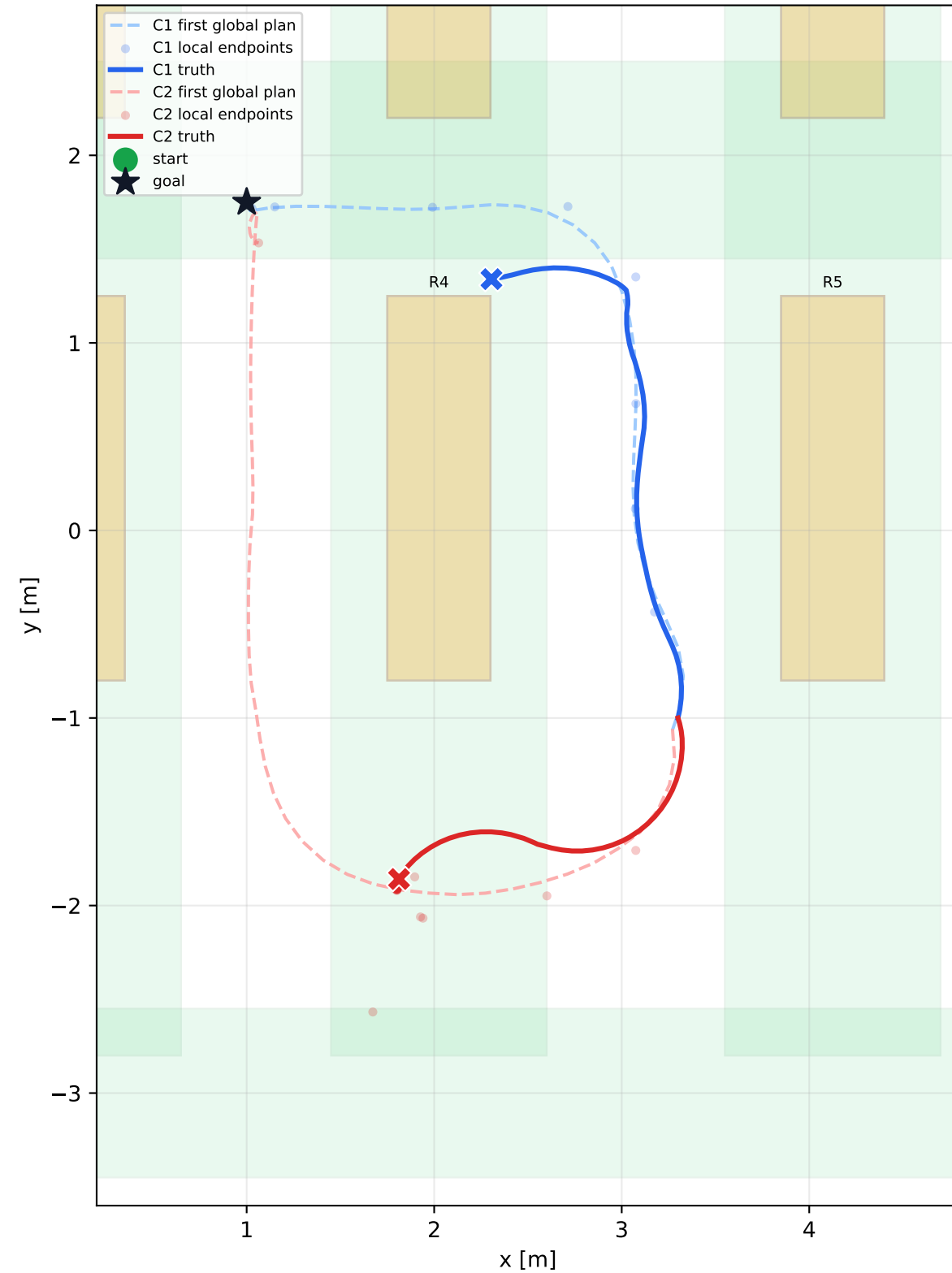


F45 - Tracking yaw + local solver timing diagnostic

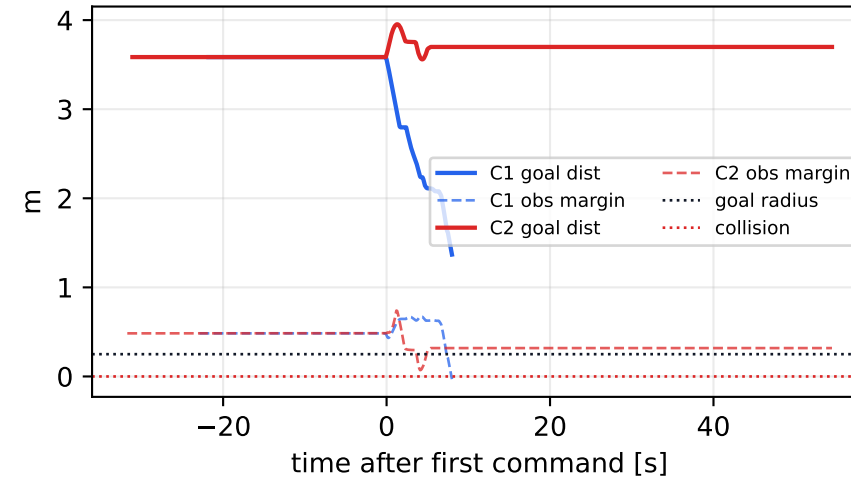
R4

R5

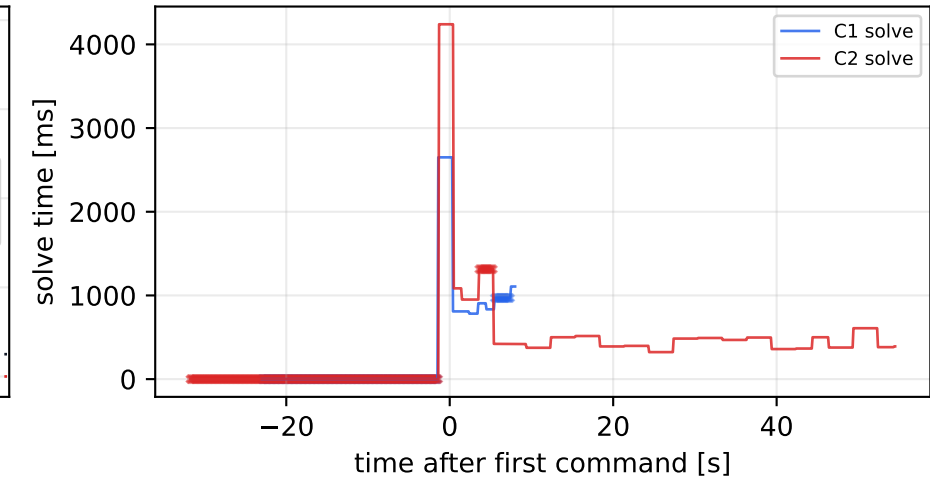
(a) First route, local endpoints, executed trajectory



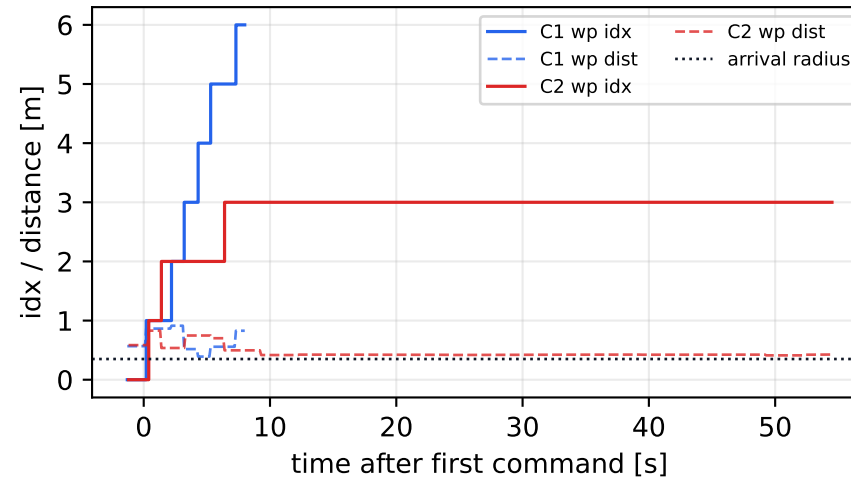
(b) Goal progress and obstacle margin



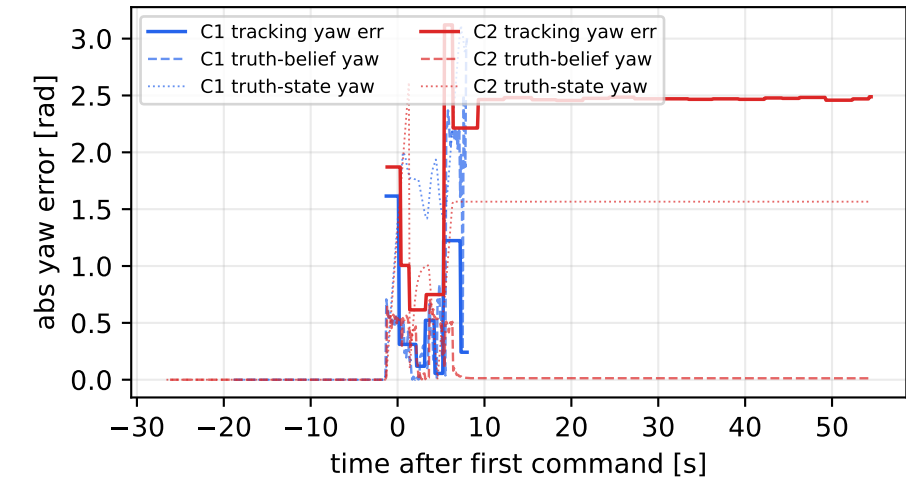
(c) Local solve time (x = non-success)



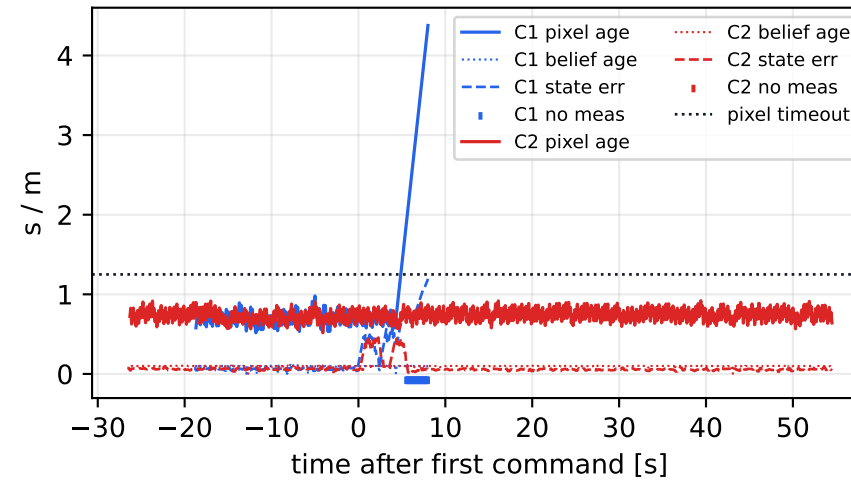
(d) Waypoint progress



(e) Heading/yaw errors



(f) Perception freshness and localization



F45 controlled tracking-yaw smoke

C1: collision, path=3.09m, min_goal=1.37m
err=0.215m, min_obs=-0.030m
solve μ /p90=1151/1724 ms, 21 ms/eval
maxiter/non-success rows=6%
yaw track μ /p90=0.68/1.62 rad
pixel age p90/max=1.71/4.38s

C2: interrupted, path=nanm, min_goal=nanm
err=nanm, min_obs=nanm
solve μ /p90=689/1058 ms, 125 ms/eval
maxiter/non-success rows=2%
yaw track μ /p90=2.30/2.48 rad
pixel age p90/max=0.85/0.92s

Interpretation:

1. Spawn yaw is clean; yaw problems are runtime/tracking, not init.
2. Global plans still choose the expected route families.
3. Local H15/maxiter20 is too budget-starved: frequent maxiter hits.
4. C1 additionally suffers stale camera belief before collision.
5. C2 localizes better but still collides: local tracking/clearance remains failing.